

## Author

HARSH SHARMA

24F2000611

24F2000611@ds.study.iitm.ac.in

I am a standalone student , currently in the last semester of diploma level , have interest in ML , DL and Backend development.

## AI/LLM usage

I have 15-20% done with the help of Gemini 2.5 pro mostly for UI improvements and suggestions , query optimization , and debugging .

## Description

The project's aim is to develop a Single Page Application for Placement Portal using Python as Backend and vue js as Frontend , which has an admin(Institute) which is pre-existing, students who can register themselves,view the ongoing drives and apply them. Companies can register themselves and post new drives after admin's approval.

## Technologies used

Python : For Backend development.

Flask Security : For session management , user authentication and role based access.

SQLAlchemy : Object relational mapper for interacting with the database using objects instead of raw SQL.

Vue & Bootstrap : for frontend development and design.

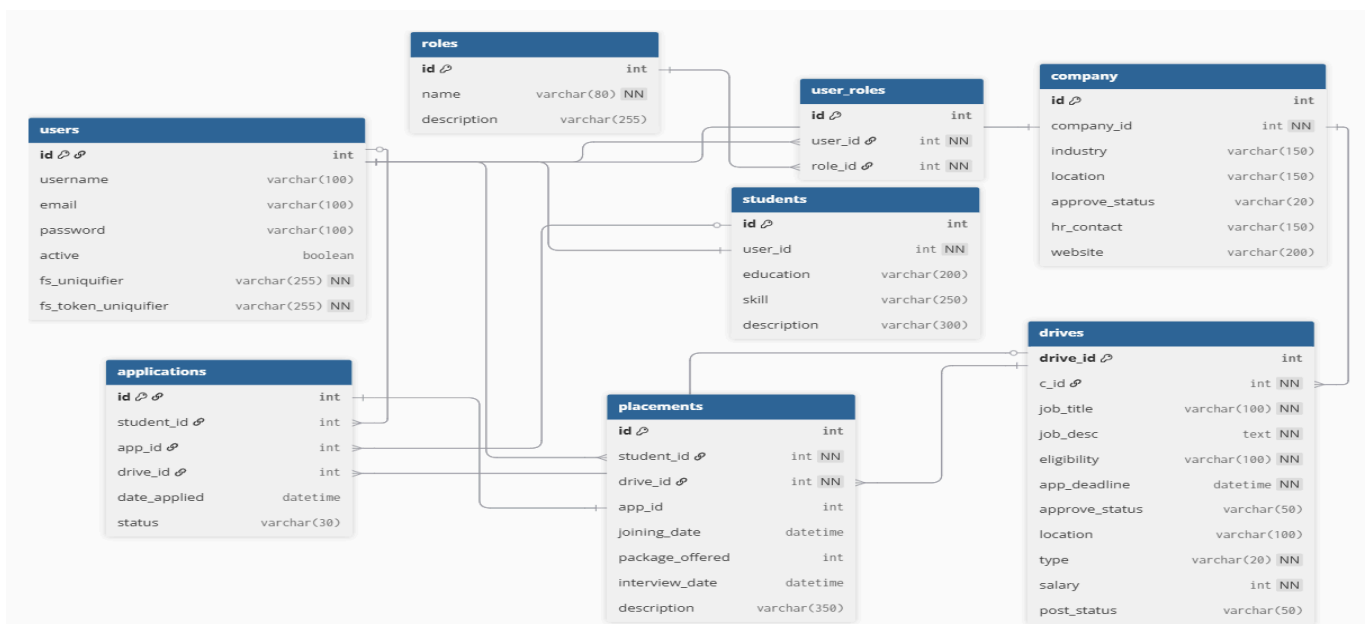
SQLITE : for database for efficient data storage.

Redis : For caching

Celery : For batch job processing

Mailhog : For receiving the messages and alerts.

## DB Schema Design



## API Design

### 1. Authentication API:

- api/login : POST request user Logging in a user
- api/logout : POST request for Logging out a user
- api/company/register : POST request for registering a company
- api/student/register : POST request for registering a student

### 2. Student API's

- api/student : GET request for fetching the student dashboard
- api/student/profile : Get request for fetching student profile
- api/student/profile/update : POST request for updating student profile
- api/student/applyjob : POST request for student to apply for jobs
- api/student/search : GET request for searching
- api/student/offer-letter : GET request for showing all the offer letters received

### 3. Admin API's

- api/admin : GET request for admin dashboard
- api/admin/approvals : POST request for approving a drive or a company
- api/admin/delete : POST request for deleting a drive , user, or application
- api/admin/search : POST request for searching drive, user, application

### 4. Company API's

- api/company : GET request for fetching company dashboard
- api/company/profile : GET request for viewing the company profile
- api/company/profile/update : POST request to update company profile
- api/company/profile/post-job : POST request to create a new drive
- api/company/delete : POST request for deleting a drive
- api/company/drive-status : POST request for updating the drive status
- api/company/drive/info/status : POST request for updating the student application status
- api/company/drive/info/<int:drive\_id> : GET request for viewing the drive details
- api/company/drive/info/offer-letter : POST request to send offer letter to selected student.

## Architecture and Features

The folder structure is as follows : Folders Backend , Frontend ,and [app.py](#) for running the application , Backend folder has all the backend design for API's , the Frontend folder has all the vue files for displaying the UI.

The features of the application are as follows :

- 1 . Admin : Pre-existing user for managing all the drives, applications , students and companies. Admin will approve the companies and their drives , view students and their applications. Delete a student, company , drive and application . The admin home page shows all the information as well as the statistics like number of students, drives, companies and applications , admin can search for students or companies or drives.
2. Student : Pre-approved User who can register and see all the ongoing placement drives , apply to them , see their status , a dedicated page for seeing the job offers received and

applications applied to . Students cannot apply for an already applied drive, receive interview notification one day before the interview date.

3. Company : Users who require admin's approval for logging in after registering , can create a drive which is visible only after admin's approval , delete the drive, view the number of students and their details . Update the status of a drive to closed or Active , update the student application status to selected , rejected , interview , send them an offer after selecting them.
4. Automatic Backend jobs like interview schedule reminders for students and monthly analytics reports for companies are emailed .

## Video

 Video.mp4